KINGDOM OF SAUDI ARABIA
IMAM MOHAMMAD IBN SAUD ISLAMIC UNIVERSITY
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
INFORMATION SYSTEMS DEPARTMENT
BACHELOR IN INFORMATION SYSTEMS

المملكة العربية السعودية
جامعة الإمام محمد بن سعود الإسلامية
كلية علوم الحاسب والمعلومات
قسم نظم المعلومات
بكالوريوس نظم المعلومات

SYLLABUS

IS1382: Fundamental of blockchain

CREDIT HOURS	3 credit hours (Lectures: 3)

PREREQUISITE	IS1250

Instructor:
Contact information and office hours
Office No: To be announced (TBA) **Office Hours: TBA** **E-mail: _____@imamu.edu.sa**

COURSE DESCRIPTION
It introduces theoretical and implementation aspects of blockchain technology, which contains all the necessary material to become a blockchain technical expert. It includes an in-depth insight into the need for decentralization, smart contracts, consensus both permissioned and permissionless, and various blockchain development frameworks, tools, and platforms such as Ethereum, Bitcoin, and Hyperledger Fabric.

	COURSE LEARNING OUTCOMES (CLOs)	Aligned SOs
1	**Knowledge and Understanding**	
1.1	Understand the blockchain technology	1(I)
1.2	Investigate the adoption and implementation of blockchain	1(P)
1.3	Understand the features of blockchain	1(P)
1.4		
1.5		
2	**Skills :**	
2.1	Interpret the requirements of a blockchain system	2(P)
2.2	Develop a blockchain system	2(A)

2.3	Apply the subject area's tools and techniques	2(A)
2.4		
2.5		
3	**Values:**	
3.1	Work effectively in groups and exercise leadership when appropriate	5(P)
3.2	Communicate effectively in oral and written form	3(P)
3.3		
3.4		
3.5		

TEACHING Strategies
Lectures Self-Learning

No	List of Topics	Contact Hours	Self-Learning
1	Introduction to Blockchain Technology	*3*	
2	Decentralization and Architecture of Blockchain Technology	*3*	
3	Cryptographic Primitives for Blockchain Development	*6*	*2*
4	Smart Contracts for Building Decentralized Applications	*3*	
5	Distributed Consensus for Permissionless Environment	*3*	
6	Mining Procedure in Distributed Consensus	*3*	
7	Distributed Consensus for Permissioned Blockchain	*3*	
8	Consensus Scalability in Blockchain Network	*3*	
9	Building Trust in Blockchain Network Using Collective Signing	*3*	*2*
10	Adoption of Blockchain in Enterprise Computing	*3*	*2*
11	Project Discussions	*3*	
12			
Total		**36**	**6**

TEXT BOOK
Blockchain Technology: From Theory to Practice (Studies in Autonomic, Data-driven and Industrial Computing) 1st ed. 2022 Edition by Sudeep Tanwar (Author)

REFERENCES

Recent Advances in Blockchain Technology: Real-World Applications (Intelligent Systems Reference Library, 237) 1st ed. 2023 Edition
by Sandeep Kumar Panda (Editor), Vaibhav Mishra (Editor), Sujata Priyambada Dash (Editor), Ashis Kumar Pani (Editor)

	Course Assessment Methods		
No	**Assessment Method**	**Due Week**	**% of Total Assessment**
1	**Quiz**	4	10
2	**Assignment**	9	10
3	**Midterm**	7	20
4	**Project / Lab Exam**	11	20
5	**Final Exam**	13	40